INCH-POUND

MIL-R-7362D
AMENDMENT 3
20 February 1990
SUPERSEDING
AMENDMENT 2
23 April 1981

MILITARY SPECIFICATION

RUBBER, SYNTHETIC, SOLID, SHEET, STRIP AND FABRICATED PARTS, SYNTHETIC OIL RESISTANT

This amendment forms a part of Military Specification MIL-R-7362D, dated 2 May 1967, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 1, 2, and 3

* Section 2, delete and substitute:

*2. APPLICABLE DOCUMENTS

*2.1 Specifications, and standards. The following specification, and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standrds (DODISS) and supplement thereto, cited in the solicitation (See 6.2).

SPECIFICATIONS

FEDERAL

NN-P-530	Plywood, Flat Panel
QQ-A-250/4	Aluminim Alloy 2024, Plate and Sheet
QQ-B-626	Brass, Leaded & Nonleaded Rod, Shapes, Forgings & Flat Product TH Finished Edges (Bar & Strip)
QQ-B-750	Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, & Structural & Special Shaped Sections
UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-T-45	Tape; Gummed Paper, Reinforced and Plain (Sealing and Securing)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

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MILITARY

MIL-P-4861	Packing, Preformed, Rubber, Packing and Packing of
MIL-H-6083	Hydraulic Fluid, Petroleum Base, for Preservation & Operations
MIL-S-18729	Steel Plate, Sheet, & Strip, Alloy 4130 Aircraft Quality
MIL-R-25897	Rubber, High-Temperature, Fluid-Resistant

STANDARDS

FEDERAL

FED-STD-791	Lubricant, Liquid Fuel & Related Products Methods of Testing

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-289	Visual Inspection Guide for Rubber Sheet Material
MIL-STD-298	Visual Inspection Guide for Rubber Extruded Goods
MIL-STD-407	Visual Inspection Guide for Rubber Molded Items
MIL-STD-413	Visual Inspection Guide for Rubber O-Rings
MS29561	Packing, Preformed, O-Ring, Synthetic Lubricant Resistant

AIR FORCE-NAVY AERONAUTICAL BULLETIN

No. 438	Age Controls of Age-Sensitive Elastomeric Items

(Unless otherwise indicted, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadephia, PA 19120-5099.)

"2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solictation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D91	Lubricating Oil Precipitation Number of
ASTM D297	Chemical Analysis of Rubber Products
ASTM D395	Compression Set of Vulcanized Rubber
ASTM D412	Tension Testing of Vulcanized Rubber
ASTM D471	Change in Properties of Elastomeric Vulcanizates Resulting from Immersion in Liquids
ASTM D573	Accelerated Aging of Vulcanized Rubber by the Oven Method
ASTM D676	Indentation of Rubber by Means of a Durometer
ASTM D1329	Evaluating Low-Temperature Characteristics of Rubber and Rubber-like Materials by a Temperature-Retraction Procedure
ASTM D1414	Tension Testing of Rubber O-Rings

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI B46.1-85 Surface Texture (Surface Roughness, Waiveness & Lay)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

"2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

* 3.1, delete and substitute:

"3.1 First Article. The synthetic rubber furnished under this specification shall be a product which has met the first article tests specified herein (see 4.3). First article tests are not necessarily required on all orders (see 6.3). If there are any changes in materials or manufacturing processes, new first article tests are required."

PAGE 4

Table I, delete and substitute:

"Table I. Thickness tolerances for sheet

Nominal thickness(inch)			tolerances (inch)
	0.031 and less		± 0.010
Over	0.031 to 0.063	inclusive	± 0.012
Over	0.063 to 0.125	inclusive	± 0.016
Over	0.125 to 0.188	inclusive	± 0.020
Over	0.188 to 0.375	inclusive	± 0.031
Over	0.375 to 0.750	inclusive	± 0.047
Over	0.750 to 1.000	inclusive	± 0.093
Over	1.000		± 10%

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Table II, after line 20, add:

Property	Type I	Type II
"Air aged - 22 hours at 257°F ±2°F		
Compression set, percent max.		40
under 0.110 inch	60	
over 0.110 inch	55	

PAGE 6

4.1, delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

"4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's over-all inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

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* 4.3, title: Delete and substitute "First article inspection"

* 4.3.2, delete and substitute:

"4.3.2 First article tests. First article tests shall consist of all the tests specified in 4.6 (see 6.3).

4.4, delete and substitute:

" 4.4, Quality conformance inspection. Sampling for inspection shall be in accordance with MIL-STD-105, level II, except where otherwise indicated. Quality Conformance tests are required for all production batches of material (see 4.4.6)."

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4.4.1.1, delete and substitute:

"4.4.1.1 Batch. A batch shall be the quantity of material compounded on a mill or mixer at one time."

Add:

"4.4.1.2 Lot. A lot shall consist of all material of the identity cured in the same production run, from the same batch, and submitted at the same time for inspection."

4.4.6 and 4.4.6.1, delete and substitute:

"4.4.6 Quality conformance tests. The following tests shall be conducted on each lot of material:

Original	Air Aged
Tensile Strength	22 hours at 257°F ±2°F
Elongation	Compression Set
Hardness	

If the items are unsuitable for use as test samples, tests shall be performed on samples of identical composition and state of cure as the end item.

"4.4.6.1 Rejection criteria. A lot shall be rejected upon the failure of any sample to meet the test requirements specified herein."

* 4.5.1, delete and substitute:

"4.5.1 Control fluid. The oil aging in this specification shall be conducted using ARM-100, or approved equal, as the control fluid. The weight percent composition of ARM-100, or approved equal is:

95.9% trimethylolpropane triheptanoate (Stauffer 704)
1.0% phenyl-alpha-naphthylamine (PANA)
1.0% dioctyldiphenylamine (Vanlube 81)
2.0% Tricresyl phosphate
0.1% 2,6-Di-tert-butyl-alpha dimthylamino-para-cresol (Ethyl 703)"

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* 4.6.1, delete and substitute:

"4.6.1 Corrosion and adhesion test

"4.6.1.1 Corrosion and adhesion. Material shall be prepared for corrosion testing by inserting sufficient quantities in a desiccator or similar humidity chamber maintained at 92 percent minimum relative humidity and room temperature for 72 hours minimum. Metallic plates of the metals listed below shall be polished to a surface roughness of 4 to 16 microinches RHR finish in accordance with ANSI B46.1-85. The edges shall also be polished to reduce the formation of edge corrosion. Plates shall be washed with precipitation naphtha, as specified in method 3101 of FED-STD-791, ASTM D91-81, or similar degreasing agent. The metals used shall be as follows:

a. Aluminum alloy: QQ-A-250/4
b. Brass: QQ-B-626, composition 2
c. Phosphor bronze: QQ-P-750
d. Steel: MIL-S-18729, condition N"

"4.6.1.2 Immersion of rubber material and metallic plates. The humidified rubber material and the metallic plates shall be immersed in type I fluid of MIL-H-6083, and drained to the drip point. The rubber material and plates shall then be so laid together in a stack that at least two rubber pieces contact each specified metal. The stack shall be held together with a pressure of 20 to 30 pounds and placed in a desiccator which is maintained at 90 to 92 percent relative humidity at room temperature. A separate set of metallic plates shall also be prepared (buffing, cleaning, and dipping in rust-preventive fluid and drained) and placed in this desiccator in such a manner that the control plates do not touch each other or any of the rubber.

NOTE: (di-potassium acid phosphate, K_2HPO_4, when placed in distilled water in sufficient quantity to produce a concentrated solution, will maintain approximately 92 percent humidity in a sealed desiccator at 68°F temperature.) Test time shall be 14 days. No more than 15 minutes should elapse between the time the test samples are removed from the prehumidifying chamber and placed in the stacked condition in the second humidity chamber.

"4.6.1.3 Inspection at the termination of the test. At the termination of this test, there shall be no adhesion of the rubber material to the metals. There shall be only slight evidence of pitting, erosion, corrosion, or bad discoloration on the plates, as shown by the following procedure, unless metallic plates are entirely unaffected by the test:

" a. The surfaces of the plates that were in contact with the packings shall be inspected for discoloration, deposits, pitting, etc. If any exists, the surfaces of the plates shall be washed in precipitation naphtha. Deposits determined as rubber compounds or elements therefrom, which can be removed by this process and which do not occur on the separate control plates, shall be construed as adhesion.

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" b. If any other marks remain on the surfaces of the plates after step(a) above, the surfaces shall be lightly polished with a nonabrasive cloth buff. Any pits or eroded marks remaining after this process shall be construed to be corrosion. Discoloration or staining (marks which do no physically affect the surface of the plates and which easily wash or buff off) shall not be considered detrimental. If any doubt should arise about the presence of pitting, erosion, or corrosion on the metal plates, a microscope of approximately 10 to 15 power magnification shall be used to determine the actual condition."

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* 4.6.4.3, line 10: Delete "Stauffer BLend 7700" and substitute "ARM-100, or approved equal"

Add:

"4.6.4.4 Air aged compression set. Compression set shall be determined in accordance with ASTM D395, Method B unless otherwise specified. Specimens for type I material shall be two o-rings. Two circular plied-up buttons with dimensions of 0.129 ± 0.010 inch diameter and approximately 0.5 inch thick shall be used for type II material. The percentage of compression employed shall be 25 percent."

Add:

"4.7 Packaging inspection. Sample packages and packs and the inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5 and the documents specified therein, or as otherwise specified in the contract or order."

Section 5, title: Delete and substitute "PACKAGING"

5.1, delete and substitute:

"5.1 Preservation - Packaging. Preservation - Packaging shall be level A or C as specified (see 6.2)."

5.1.1.2.2, line 5: Delete "NN-P-515" and substitute "NN-P-530"".

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5.2.1.2, line 11: Delete "NN-P-515" and substitute "NN-P-530"

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6.2.h, delete and substitute:

"h. Applicable levels of preservation - packaging and packing (see Section 5)."

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* 6.3, delete and substitute:

"6.3 First article. When first article inspection is required, the contracting officer shall provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first one production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation."

* 6.4, delete and substitute:

"6.4 Source of fluid. ARM-100 may be obtained from SAE, Attn: Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096."

Add:

"6.5 Changes from previous issue. The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

Custodian:
Army - MR
Navy - AS
Air Force - 11

Review activities
Army - MI, AR
Air Force - 99
DLA - IS

Preparing activity:
Air Force - 11

Project Number: 5330-0782

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